

HARSHIL KARELIYA

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PROFESSIONAL SUMMARY

Data Science professional with hands-on experience in machine learning, deep learning, and computer vision. Skilled in Python, TensorFlow, Keras, and transfer learning with pretrained models (ResNet50, DenseNet121). Built end-to-end AI solutions including medical image classification, RAG pipelines, and predictive analytics systems. IABAC-certified with proven ability to deliver production-ready models from data preprocessing through deployment.

TECHNICAL SKILLS

Languages: Python, SQL
Libraries: Pandas, NumPy, Scikit-learn, TensorFlow, Keras, OpenCV
Visualization: Matplotlib, Seaborn, Power BI
Core Competencies: Machine Learning, Deep Learning, Computer Vision, Statistics, Data Preprocessing, RAG
Tools & Platforms: Git, GitHub, Jupyter Notebook, VS Code, Excel

WORK EXPERIENCE

AI/ML Intern | Rubixe – AI Solutions Company

June -2025 to Present

- Developed 10+ AI/ML projects across business and research domains using Python, Scikit-learn, Pandas, and NumPy.
- Built and optimized Machine Learning, Deep Learning (CNNs, Transfer Learning), and YOLO models for classification, regression, image classification, and object detection.
- Developed an LLM-powered RAG chatbot using NLP, embeddings, vector databases, and Large Language Models for document-based question answering.
- Performed data preprocessing, feature engineering, hyperparameter tuning, and model evaluation to improve model performance.

PROJECTS

AI-Powered RAG Document Chatbot

- Built a local AI-powered document question-answering system using **LangChain**, **Google Gemini 3.5 Flash**, **FAISS**, and **HuggingFace embeddings**. Implemented document ingestion, semantic retrieval, sliding-window conversation memory, and citation-based responses with a customizable Streamlit UI.

AI Automation Workflows using n8n

- Built an AI-powered personal assistant using Streamlit and n8n that automates calendar, email, task management, and expense tracking through natural language interactions. Integrated Google Workspace APIs, Gemini AI, and SerpApi to deliver real-time productivity automation.

Indian Sign Language Recognition

- Built an Indian Sign Language recognition pipeline using ResNet50 transfer learning with ImageNet pretraining. Implemented data augmentation, EarlyStopping, and ModelCheckpoint callbacks. Achieved 99.46% test accuracy across 24 gesture classes.

Pneumonia X-Ray Classification

- Developed binary chest X-ray classification system (Normal vs Pneumonia) using transfer learning with DenseNet121 pretrained on ImageNet. Trained and compared four models, Applied data augmentation and callbacks. Achieved **91.51% test accuracy** and **97.69% recall**.

EDUCATION

Bachelor of Computer Applications (BCA)

Saurashtra University, Rajkot

Data Science Specialization

Datamites, Ahmedabad

CERTIFICATIONS

Data Science Certification – Datamites

Certified Data Scientist – IABAC (International Association of Business Analytics Certifications)

Certified Data Scientist – NASSCOM